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 Studierichting: Informatica
 Oefeningenreeks 5D nr. 14

$$y^2 = x^4 + x^5$$

$$\Leftrightarrow y = \sqrt{x^4 + x^5}$$

$$\Leftrightarrow y = x^2\sqrt{x+1}$$

$$\int_{-1}^0 x^2\sqrt{x+1}dx$$

$$t = x + 1 \Leftrightarrow x = t - 1$$

$$x = -1 \Rightarrow t = 0, x = 0 \Rightarrow t = 1$$

$$= \int_0^1 (t-1)^2 \cdot \sqrt{t} dt$$

$$= \int_0^1 (t^2 - 2t + 1) \cdot \sqrt{t} dt$$

$$= \int_0^1 \left(t^{\frac{5}{2}} - 2t^{\frac{3}{2}} + \sqrt{t} \right) dt$$

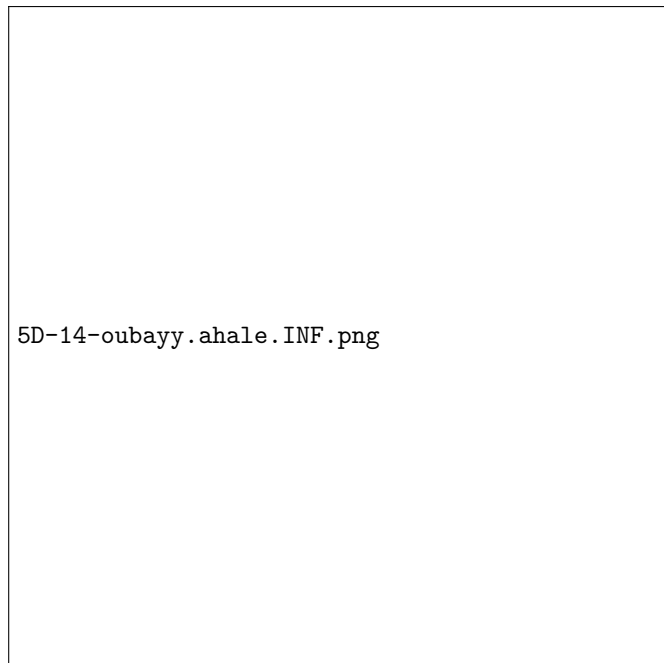
$$= \int_0^1 t^{\frac{5}{2}} dt - \int_0^1 2t^{\frac{3}{2}} dt + \int_0^1 \sqrt{t} dt$$

$$= \left[\frac{2t^{\frac{7}{2}}}{7} \right]_0^1 - \left[\frac{4t^{\frac{5}{2}}}{5} \right]_0^1 + \left[\frac{2t^{\frac{3}{2}}}{3} \right]_0^1$$

$$= \frac{16}{105}$$

Voor de volledige oppervlakte vermenigvuldigen we met 2

$$= \frac{32}{105}$$



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